hovid Co-Trimoxazole Tablet B.P. 960 mg

VISUP01-M

DESCRIPTION
Oblong, white uncoated tablet, bevel-edged, shallow convex faces, break-bar embossed on one face.

COMPOSITION
Sulphamethoxazole 800 mg and trimethoprim 160 mg/tablet.

ACTIONS AND PHARMACOLOGY
Trimethoprim is an antibacterial.

Sulfamethoxazole is a sulphonamide.

Sulfamethoxazole competitively inhibits the utilisation of para-aminobenzoic acid in the synthesis of dihydrofolate by the bacterial cell resulting in bacterio stasis. Trimethoprim reversibly inhibits bacterial dihydrofolate reductase (DHFR), an enzyme active in the folate metabolic pathway converting dihydrofolate to tetrahydrofolate. Depending on the conditions the effect may bebactericidal. Thus trimethoprim and sulfamethoxazole block two consecutive steps in the biosynthesis of purines and therefore nucleic acids essential to many bacteria. This action produces marked potentiation of activity in vitrobetween the two agents.

Trimethoprim binds to plasmodial DHFR but less tightly than to the bacterial enzyme. Its affinity for mammmalian DHFR is some 50,000 times than for the corresponding bacterial enzyme.

Many of common pathogenic bacteria are sensitive in vitro to trimethoprim and sulfamethoxazole at concentrations well below those reached in blood, tissue fluids and urine after administration of recommended doses. In common with other antibiotic, however, in vitro activity does not necessarily imply that clinical efficacy has been demonstrated and it must be noted that satisfactory sensitivity testing is achieved only with recommended media free from inhibitory substances especially thymidine and thymine.

PHARMACOKINETICS
Trimethoprim is readily absorbed from the gastrointestinal tract and peak concentrations in the circulation occur between 1 and 4 hours after a dose is taken. Trimethoprim is a weak base with a pKa of 7.4, it is lipophilic. Tissue levels of trimethoprim are generally higher than corresponding plasma levels, the lungs and kidney showing especially high concentrations. About 50% is bound to plasma proteins. The elimination half-life is in the range of 8.6 - 17 hours in the prescence of normal of normal renal function. It is increased by a factor of 1.5 to 3.0 when the creatinine clearance is less than 10ml/min. There appears no significant difference in the elderly compared with the young patients. About 40 - 60% of a dose is excreted unchanged in the urine within 24 hours, together with metabolites. Trimethoprim is removed by haemodialysis to some extent.

Sulfamethoxazole is readily absorbed from the gastrointestinal tract and peak plasma concentrations are reached between 1 and 4 hours. Sulfamethoxazole is a weak acid with a pKa of 6.0. The concentration of active sulfamethoxazole in a variety of body fluids is of the order of 20 to 50% of the plasma concentration. About 66% is bound to plasma albumin and the plasma half-life is in the range of 9 - 11 hours. It is prolonged in patients with severe renal impairment. About 15% of sulfamethoxazole in the blood is present as the acetyl derivative. Elimination in the urine is dependent on pH. In the region of 25% of a single 2g dose of sulfamethoxazole has been reported to be excreted in the urine within eight hours, around 60% being in the form of the acetyl derivative.

When Co-trimoxazole is administered, plasma concentrations of trimethoprim and sulfamethoxazole are generally in the ratio of 1:20; in urine this ratio may vary from 1:1 to 1:5. About 50% of each drug is excreted in the urine within 24 hours, but a larger proportion of sulfamethoxazole appears as inactive metabolite.

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INDICATIONS
For treatment of :
- Urinary tract infections caused by *E. coli, Klebsiella, Enterobacter, P. mirabilis, P. vulgaris* and *Morganella morganii.*
- Acute otitis media in children caused by *H. influenzae* and *Streptococcus pneumoniae.*
- Acute exacerbations of chronic bronchitis in adults caused by *H. influenzae* and *Streptococcus pneumoniae.*
- Enteritis caused by *Shigella flexneri* and *S. sonnei.*
- Pneumonitis caused by *Pneumocystis carinii.*

CONTRAINDICATIONS
- Avoid in patients known to be hypersensitive to sulphonamides, frusemide, sulfones, thiazide diuretics, sulfonylureas or carbonic anhydrase inhibitors.
- Not recommended in patients with megaloblastic anaemia.
- Not recommended during pregnancy and lactation.

PRECAUTIONS
- Caution in patients with glucose-6-phosphate dehydrogenase (G6PD) deficiency, porphyria, renal and hepatic function impairment and in nursing mothers.
- All patients receiving prolonged treatment should be given regular blood examinations.
- Adequate fluid intake will help to prevent risk of crystalluria.
- There is an increased risk of adverse reactions in elderly patients in particular blood disorders and adverse renal/hepatic reactions.

WARNING
Fatalities associated with the administration of sulphonamides and trimethoprim, either alone or in combination, have occurred due to severe reactions, including Stevens-Johnson syndrome, toxic epidermal necrolysis and other reactions. The drug should be discontinued at the first appearance of skin rash or any sign of adverse reactions.

PREGNANCY AND LACTATION
Pregnancy:
Co-trimoxazole should not be used in pregnancy as the safety in pregnancy has not been established. Co-trimoxazole interferes with folate metabolism and can cause teratogenic effects if given in the first trimester.

Co-trimoxazole can cause neonatal haemoylosis and methaemoglobinaemia when used in the third trimester, if given close to delivery kernicterus may occur due to displacement of bilirubin. Other toxicities that may be observed in the new born include jaundice and haemalytic anaemia. The risk of kernicterus is higher in infants at increased risk of hyperbilirubinaemia, such as if the infant is ill, stressed or premature or has glucose-6-phosphate dehydrogenase deficiency.

Lactation:
Co-Trimoxazole appears in breast milk in negligible amounts and the risk appears to be low. However, there is a risk of kernicterus if the infant is at increased risk of hyperbilirubinaemia.

MAIN SIDE/ADVERSE EFFECTS
Increased sensitivity of skin to sunlight, itching, headache, unusual taste in mouth, Stevens-Johnson syndrome, Lyell's syndrome, blood dyscrasias, hypersensitivity, hepatitis, methemoglobinemia, sore throat, bluish fingernails, lips and skin, blood in urine, lower back pain, crystalluria and thyroid function disturbance.

DRUG INTERACTIONS
- **Concurrent therapy with the following drugs is not recommended:**
  Aminobenzoic acid (PABA), methenamine, penicillins.
- **Concurrent therapy with the following drugs may necessitate dosage adjustments:**
  Oral anticoagulants, oral hypoglycaemics, methotrexate, phenytoin, thiopental and sulfinpyrazone.

OVERDOSAGE
Clinical features: Crystalluria, oliguria, anuria, bone marrow depression, gastrointestinal irritation and fatigue.
Treat overdosage by emesis or gastric lavage, if appropriate. Oral sodium bicarbonate and increasing fluid intake may help to eliminate sulphamethoxazole in the urine. Bone marrow depression due to trimethoprim may be treated with calcium folinate 3 - 6 mg IV, then 15 mg daily by mouth for 5 - 7 days.

DOSAGE AND ADMINISTRATION
This medication should preferably be taken with a full glass (240 ml) of water or on an empty stomach (either 1 hour before or 2 hours after meals).
Fluid intake should be sufficient to maintain urine output of at least 1200 to 1500 ml per day in adults.
Therapy should be continued for at least 10 to 14 days in acute exacerbation of chronic bronchitis; for 7 to 10 days in urinary tract infections; 5 days in shigellosis; 10 days in acute otitis media in children; 14 days in Pneumocystis carinii pneumonitis.

Adults : Antibacterial systemic
Oral, 800 mg of sulphamethoxazole and 160 mg of trimethoprim every twelve hours, or as directed.

Children : Antibacterial systemic
Infants and children up to 40 kg of body weight :
Oral, 20 mg of sulphamethoxazole and 4 mg of trimethoprim per kg of body weight every twelve hours; or as directed.
Children 40 kg of body weight and over : See adult dose.

Note: Infants under 2 months of age - Use is contraindicated since sulfonamides may cause kernicterus in neonates.

Storage	: Store below 25°C. Protect from light and moisture.
Presentation/Packing	: Blisters of 10 x 10's.
Product Registration Holder	: HOVID Bhd.,121, Jalan Tunku Abdul Rahman, 30010 Ipoh, Malaysia.
Manufactured by	: HOVID Bhd., Lot 56442, 7 1/2 Miles,Jalan Ipoh/Chemor, 31200 Chemor, Malaysia.
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